

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I Gurusurya.G192311332_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Gurusurya

Signature of the student:

Annexure A

Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

- A. Explain why this issue occurs.
- B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- A. Identify the most suitable algorithm for this scenario.
- B. Justify your choice based on security strength and efficiency.

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

- A. Explain how the key exchange works in this scenario.
- B. Identify a possible security threat and suggest a solution to mitigate it.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

- A. Describe how RSA can be used for encryption and decryption in this scenario.
- B. Explain how key distribution and management play a role in maintaining security.

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

- A. Analyze the constraints of IoT devices.
- B. Recommend a suitable cryptographic approach and justify your answer.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Secure Data Transmission using Block Cipher

A. Why does ECB mode reveal patterns?

ECB (Electronic Codebook) mode encrypts each plaintext block independently using the same key.

- Identical plaintext blocks produce identical ciphertext blocks.
- If the same financial data (such as account numbers or transaction records) appears repeatedly, the encrypted output will also repeat.
- An attacker cannot directly read the data but can recognize patterns, which may reveal useful information about the message structure.

Example:

Plaintext ECB Ciphertext

Block A Cipher X

Block A Cipher X

Block B Cipher Y

Block A Cipher X

Repeated plaintext produces repeated ciphertext, making patterns visible.

B. Suggest a more secure block cipher mode

Recommended Mode: CBC (Cipher Block Chaining)

Reason:

- Each plaintext block is XORed with the previous ciphertext block before encryption.
- The first block uses an **Initialization Vector (IV)**.
- Even if two plaintext blocks are identical, their ciphertext blocks will be different because each block depends on the previous ciphertext.

Advantages of CBC:

- Hides plaintext patterns.
- Provides better confidentiality than ECB.
- Suitable for transmitting sensitive financial information.

Conclusion:

CBC mode is more secure because it prevents attackers from identifying repeated data patterns in encrypted financial transactions.

2. Choosing an Encryption Algorithm for Banking System

A. Most suitable algorithm

Recommended Algorithm: Blowfish

B. Justification

Algorithm Security

DES Weak (56-bit key)

3-DES Strong

Blowfish Strong (up to 448-bit key)

Performance Suitability

Fast Not recommended

Slow Secure but inefficient

Fast Best choice

Reasons:

- DES is vulnerable to brute-force attacks due to its small key size.
- 3-DES provides better security but is approximately three times slower than DES.
- Blowfish offers strong encryption with better performance, making it suitable for banking applications that require secure and efficient transaction processing.

Conclusion:

Blowfish provides an excellent balance between strong security and high efficiency for banking systems.

3. Key Exchange in an Unsecured Network

A. How Diffie–Hellman key exchange works

1. Both users publicly agree on:
 - A prime number (**p**)
 - A generator (**g**)
2. User A selects a private key (**a**).
3. User B selects a private key (**b**).

4. User A computes:

$$A = g^a p$$

and sends the public key to User B.

5. User B computes:

$$B = g^b p$$

and sends the public key to User A.

6. User A computes:

$$K = B^a p$$

7. User B computes:

$$K = A^b p$$

Both users obtain the same shared secret key without transmitting it over the network.

B. Security threat and solution

Threat: Man-in-the-Middle (MITM) Attack

An attacker intercepts the exchanged public keys and replaces them with fake public keys.

As a result:

- User A establishes a secret key with the attacker.
- User B also establishes a secret key with the attacker.
- The attacker can read, modify, and forward messages without either user realizing it.

Solution:

- Use **Digital Signatures**.
- Use **Digital Certificates (PKI)** to authenticate public keys.
- Implement authenticated protocols such as **TLS**, which combines Diffie–Hellman with authentication.

Conclusion:

Authentication mechanisms prevent MITM attacks and ensure that the exchanged public keys belong to the intended users.

4. Public Key Cryptography for Email Security

A. How RSA is used for encryption and decryption

RSA uses a pair of keys:

- **Public Key** – shared with everyone.
- **Private Key** – kept secret by the owner.

Encryption Process:

1. The sender obtains the recipient's public key.
2. The email is encrypted using the recipient's public key.
3. The encrypted email is sent over the network.

Decryption Process:

1. The recipient receives the encrypted email.
2. The recipient uses their private key to decrypt the message.
3. Only the intended recipient can read the email.

This ensures confidentiality during email communication.

B. Role of key distribution and management

Proper key management is essential for maintaining RSA security.

Key Distribution

- Public keys must be distributed securely.
- Digital certificates issued by trusted Certificate Authorities (CAs) verify the authenticity of public keys.

Key Management

- Keep private keys confidential.
- Rotate keys periodically.
- Revoke compromised keys immediately.
- Store keys securely using hardware or secure software solutions.

Conclusion:

Effective key distribution and management prevent impersonation, unauthorized access, and key compromise.

5. Cryptography for IoT Devices

A. Constraints of IoT devices

IoT devices typically have limited resources:

- Low processing power
- Limited memory
- Limited storage
- Low battery capacity
- Restricted communication bandwidth

Because of these limitations, cryptographic algorithms should require minimal computation and memory while maintaining strong security.

B. Recommended cryptographic approach

Recommended Algorithm: Elliptic Curve Cryptography (ECC)

Justification:

Feature	RSA	ECC
Key Size	Large (1024 bits or more)	Small (160 bits provides similar security)
Computation	Higher	Lower
Memory Usage	Higher	Lower
Power Consumption	Higher	Lower
IoT Suitability	Less suitable	Highly suitable

Advantages of ECC

- Strong security with much smaller keys.
- Faster cryptographic operations on resource-constrained devices.
- Lower memory and storage requirements.
- Reduced power consumption, extending battery life.
- Well suited for embedded and smart home IoT systems.

Conclusion:

ECC is the preferred cryptographic solution for IoT devices because it provides equivalent security to RSA with significantly lower computational, memory, and power requirements, making it ideal for low-power smart home environments.